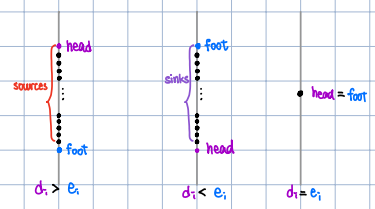
Def: A den is a tuple $(h, p, \vec{d}, \vec{e})$ where $h \in \mathbb{Z}_+$, $p \in \mathbb{R}\setminus\mathbb{Q}$, $\vec{d} = (d_0, \dots, d_h)$, $\vec{e} = (e_0, \dots, e_h) \in \mathbb{Z}^{h+1}$ satisfying

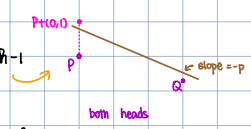
- $\dfrac{d_i - d_j + 1}{j - i} > p \quad \forall\ 0 \le i < j \le h-1$
- $\dfrac{e_i - e_j - 1}{j - i} < p \quad \forall\ 1 \le i < j \le h$
- $d_0 > e_0$, $d_h < e_h$, $\sum_{i=0}^{h} d_i = \sum_{i=0}^{h} e_i$.

Def: We call
- (i, d_i) : heads
- (j, e_j) : feet
- (i, j) s.t. $e_i < j \le d_i$: sources
- (i, j) s.t. $d_i < j \le e_i$: sinks

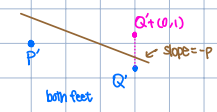


Set $\vec{g} = (g_0, \dots, g_h)$ where $g_k = \sum_{i=0}^{k-1} (d_i - e_i)$.

- $\dfrac{d_i - d_j + 1}{j - i} > p \quad \forall\ 0 \le i < j \le h-1$



- $\dfrac{e_i - e_j - 1}{j - i} < p \quad \forall\ 1 \le i < j \le h$



- $d_0 > e_0$, $d_h < e_h$, $\sum_{i=0}^{h} d_i = \sum_{i=0}^{h} e_i$

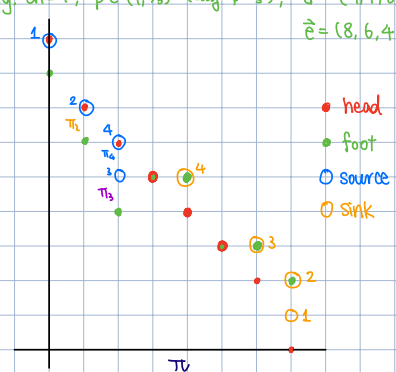
at least one source on $x=0$ — at least one sink on $x=h$ — # sources = # sinks

Note that $\dfrac{d_i - d_j + 1}{j - i} > p > \dfrac{e_i - e_j - 1}{j - i} \quad \forall\ 1 \le i < j \le h-1 \Rightarrow d_i - e_i + 1 \ge d_j - e_j \quad \forall\ 1 \le i < j \le h-1$.

If there is a sink on $x=i$ and a source on $x=j$, then $d_i - e_i < 0$ and $d_j - e_j > 0 \Rightarrow d_i - e_i + 1 \le 0 < d_j - e_j$

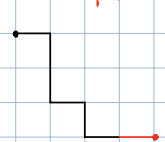
i.e. No source and sink can be on the same vertical line, and all sources are strictly left to all sinks.

e.g. $h=7$, $p \in (1, 8/5)$ (say $p = \frac{7}{5}$), $\vec{d} = (9,7,6,5,4,3,2,0)$, $\vec{e} = (8,6,4,5,5,3,3,2)$ $\Rightarrow$ $\vec{d} - \vec{e} = (1,1,2,0,-1,0,-1,2)$, $\vec{g} = (1,2,4,4,3,3,2)$



Def: An E-end path is a S/E lattice path that ends with an east step.

e.g.



Def: Let π_1, π_2 be E-end paths. π_1 and π_2 are nested with π_1 below π_2 if

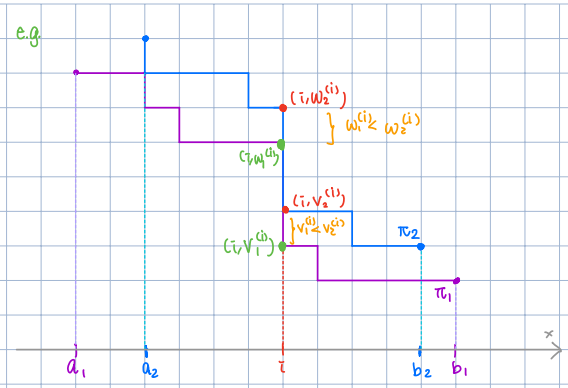
(1) the x-coordinates $[a_2, b_2]$ of points of π_2 is contained in x-coordinates $[a_1, b_1]$ of points of π_1;

(2) for every integer $i \in [a_1, b_1]$, intervals $[v_1^{(i)}, w_1^{(i)}] := \pi_1 \cap (x=i)$ (y-coordinates of π_1 on $x=i$) and $[v_2^{(i)}, w_2^{(i)}] = \pi_2 \cap (x=i)$ (y-coordinates of π_2 on $x=i$) satisfy $v_1^{(i)} < v_2^{(i)}$, $w_1^{(i)} < w_2^{(i)}$

* Nested paths can share south steps but not east steps.

* Nesting is transitive: π nested below π', π' nested below $\pi'' \Rightarrow \pi$ nested below π''.

e.g.



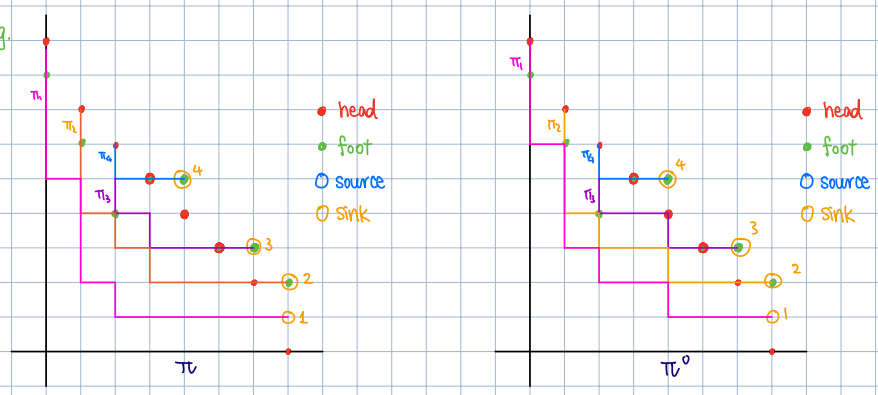
Def: A nest in a den is a system of nested E-end paths $\{\pi_1, \pi_2, \dots, \pi_\ell\}$ connecting the sources to the sinks, such that every lattice point on any path (except the sink at the end, i.e. (h, d_h)) lies weakly below the heads on its vertical line.

lexicographic order by (x,y) (bottom to top, left to right:)

order lexicographically by $(-x,y)$ (bottom to top, right to left)

i.e. non-sink lattice points lie weakly below the heads, and π_k nested below π_ℓ for $k<\ell$

e.g.



Remark:

- g_k counts sources minus sinks with x-coord $\le k$ $\Rightarrow$ g_k = # paths π_j in π that have an E-step from $x=k-1$ to $x=k$
  = # paths π_j in π that have a non-sink lattice point on $x=k-1$
  must be the first g_k paths $\pi_1, \dots, \pi_{g_k}$ because if π_j has an E-step from $x=k-1$ to $x=k$, so does π_i nested below π_j (i.e. $i<j$.)
- Since all sources are left of all sinks and there is at least one source on $x=0$ and at least one sink on $x=h$, $\vec{g}$ is positive and unimodal, with max = # sources (= # sinks)
  i.e. $g_i > 0 \ \forall\, 1 \le i \le h$
- It is possible to have an abandoned den with no nest (e.g. some source is lower than the matching sink).